

Kaya Omer

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EDUCATION & AWARDS

University of Toronto

Bachelor of Science in Computer Science, Focus in Artificial Intelligence

Toronto, ON

Sep. 2023 – May 2028

President's Scholar of Excellence

Awarded to the most highly qualified undergraduate students.

Associated with University of Toronto

March 2023

Janet Faulds and M. Ann Marshall Family Award.

Awarded to the qualified students at St Michael's College.

St. Michael's College at University of Toronto

Nov 2023

EXPERIENCE

Agentic AI Researcher Intern

The Matter Lab

May 2026 – Present

Toronto, ON

- Engineering specialized AI agents for **quantum chemistry** workflows using **PySCF** and **ORCA** to automate electronic structure calculations, molecular property computation, and API interfacing.
- Architecting autonomous LLM-driven scientific discovery pipelines ("**El Agente**"), building agentic workflows that orchestrate quantum chemistry tools, external APIs, and data retrieval systems.
- Developing an AI-native framework integrating LLM-based agents with internal web interfaces, enabling automated execution of complex computational chemistry protocols.

Machine Learning Research Assistant

CECCS

April 2026 – August 2026

Toronto, ON

- Modernized legacy ML classification pipeline using **Python**, **scikit-learn**, and **NLP** (word2vec, embeddings), retraining **Random Forest** models to fix silent feature-loss bugs.
- Architected fault-tolerant, automated **LLM evaluation & inference pipelines** featuring structured output enforcement, exponential back-off, and cross-model agreement validation, reducing target false positives by 4x.

Software Engineer Intern

CIBC

Jan 2026 – April 2026

Toronto, ON

- Engineered internal AI integrations to automate workflows; built data pipelines using Python, Apache Spark, and Databricks to process large-scale financial data.
- Implemented MLOps workflows using Azure AI Studio, Azure ML, and Azure DevOps, including CI/CD pipelines for automated model deployment.

Research Assistant / Software Engineer

Acceleration Consortium

Sep 2025 – Dec 2025

Toronto, ON

- Engineered a full Computer Vision pipeline for a Self-Driving Lab; implemented Structure from Motion (SfM) using OpenCV for 3D reconstruction and dimensional accuracy analysis with NumPy.
- Developed a Robotic Data Acquisition System integrating a PiCam and stepper motor for automated 360° object scanning within a closed-loop autonomous experimentation framework.

PROJECTS

Quantum Chemistry Agent Pipeline | *Python, PySCF, ORCA, LLMs*

May 2026

- Built LLM-driven agentic workflows to automate quantum-chemistry calculations via PySCF and ORCA, orchestrating multi-step electronic structure computations with tool-use and automated error handling.

3D Reconstruction & Automation Pipeline for SDL | *Python, OpenCV, NumPy, RPi*

Sep 2025

- Engineered a CV pipeline for a Self-Driving Lab combining SfM-based 3D reconstruction with automated robotic scanning for closed-loop material quality assessment.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, Swift, JavaScript, HTML/CSS, R, Assembly

Frameworks: PyTorch, TensorFlow, React, Flask, FastAPI

Developer Tools: Git, Docker, CUDA, GCP, Azure AI Studio, Databricks, Linux

Libraries: PySCF, SciPy, OpenCV, pandas, NumPy, Matplotlib, ORCA

AI/ML: LLM Agents, Agentic Workflows, Quantum Chemistry Automation, MLOps, Computer Vision